João Barão

jrspsbarao@gmail.com | +351 925 527 576 | Coimbra, PT | LinkedIn: joãorafaelbarão | Github: JoaoBarao18

Education

Instituto Superior de Engenharia de Coimbra (ISEC) — Coimbra, Portugal
Bachelor of Science in Computer Science and Engineering — *Expected Graduation, June 2027*

- **Concentrations:** Full-Stack & Mobile Software Engineering, Distributed Systems, and Enterprise Software Architecture
- **Core Engineering Foundations:** Object-Oriented Architecture & Design Patterns, Concurrent & Low-Level Systems Programming, Modern Web App Development (React.js), Relational Database Architecture, RESTful Microservices, Cross-Platform Mobile Engineering, Data Structures & Algorithms

Technical Projects

Saber+ – AI Academic Assessment Platform | *React.js, Vite, Spring Boot, REST API, GitLab* — Oct 2026 – Present

- Co-engineering a **22-feature** web platform using **React.js** and **Spring Boot** to centralize exam banks and deliver automated, criterion-by-criterion open-answer evaluations via **LLM** integration.
- Enforced code quality across a **5-member Scrum** team over **5 sprints** using a **4-stage GitLab** pipeline (`feature` → `dev` → `qa` → `main`), Architecture Decision Records (**ADRs**), and mandatory Peer Reviews.

Deep Sea Mining – Multi-Phase Strategy Desktop Game | *Java, JavaFX, JUnit, Design Patterns* — 2026

- Architected a **4-phase** strategy game in **Java/JavaFX** using **5 design patterns** (**Facade, State, Factory, Singleton, Multiton**) to strictly decouple domain logic from the GUI and manage **3 autonomous drones**.
- Utilized asynchronous property-change observers, binary serialization, and **JUnit** test suites (minimum **2 test cases per core class**) to guarantee thread-safe UI rendering and state persistence.

ENTREGA-SO – Concurrent Delivery Dispatch System | *C, Linux/Unix, IPC, Named Pipes, Signals* — Oct 2026 – Present

- Built a **4-process** distributed system in **C (Linux)** using POSIX system calls, **Named Pipes (FIFOs)**, and anonymous pipes to orchestrate real-time order pricing and dispatch for up to **20 concurrent couriers**.
- Engineered non-blocking deadline expiration and **10-second** dead-client heartbeat detection using **Unix Signals** (`SIGINT`, `SIGTERM`) and mutex synchronization to eliminate race conditions and zombie processes.

myMenu – AI Smart Recipe Generator | *TypeScript, Supabase, Gemini API* — 2026

- Developed a full-stack web application integrating the **Google Gemini Flash LLM** with a **Supabase** backend (PostgreSQL + Google OAuth) to dynamically generate personalized recipes based on pantry inventory and dietary restrictions.

Leadership & Extracurricular Activities

Main Developer & UI/UX Designer – ShiftAppens Hackathon — Coimbra, Portugal
myMenu Project Pitch | *Figma, UI/UX, Agile Development* — *2026*

- Led front-end architecture and UI/UX design under a strict **48-hour** deadline, pitching a functional web application focused on reducing household food waste to technical judges.

Volleyball Training Assistant & Team Support (U14 Boys) — Portimão, Portugal
Agrupamento de Escolas Poeta António Aleixo — *2022 – 2024*

- Assisted the head coach during training sessions and matches, mentoring young athletes while playing competitively and fostering team discipline and communication under pressure.

Embedded Systems Trainee – Ciência Viva Course — Portimão, Portugal
Smart Systems & Microcontrollers | *Embedded C, Hardware Sensors* — *2022*

- Programmed microcontrollers in **C** to interface with physical sensors and actuators, building automated hardware-software prototypes for real-world environments.

Technical Skills

Languages: Java, JavaScript (ES6+), TypeScript, Kotlin, Dart, C++, C (System & Embedded), C#, SQL, HTML5, CSS3, MATLAB
Frontend, Mobile & UI: React.js, Vite, Flutter, Android (Kotlin), JavaFX, Responsive Web Design, Figma (UI/UX)
Backend & Databases: Spring Boot, Node.js, Express.js, RESTful APIs, Supabase (PostgreSQL, OAuth), MySQL
DevOps & Practices: Git, GitLab, GitHub, Docker, Linux/Unix, JUnit, Agile/Scrum, CI/CD Workflows, OOP Design Patterns, UML

Additional Experience

Food Service Associate (Summer Roles) — Portimão, Portugal
QB Concept (Jul–Sep 2023) & KFC (Jun–Sep 2022) — *Summers 2022 & 2023*

- Delivered direct **customer-facing service** in high-volume hospitality environments during peak summer seasons, resolving client requests swiftly in both Portuguese and English.

Languages

Portuguese: Native | **English**: C2 (Proficient)